

Operating Procedure for QB3 CNT Chips Temescal FC 1800 Evaporator Version 2025 October 8



I. Purpose

This Standard Operating Procedure (SOP) outlines requirements to be considered by an authorized user of the TEMESCAL Thermal Evaporator as well as describes the normal operation of the evaporator and any hazards that may be encountered during normal operation. Finally, the SOP explains how to minimize any hazards and how to respond in an emergency situation. This document is to be reviewed one year from the date of approval or as conditions warrant, whichever is the shorter time period.

II. Personnel

- A. Authorized Personnel: The E-Beam Evaporator may be operated only by authorized personnel who are fully cognizant of all safety issues involved in the operation of such a device. These personnel are to ensure that the E-Beam Evaporator is only operated in the manner laid out in this document. To become an authorized user, one must:
1. Complete Environment, Health & Safety (EH&S) Lab Safety class.
 2. Take the baseline BNC Safety Orientation class
 3. Read and fully understand the SOP
 4. Receive training on the Evaporator by an authorized user.
 5. Sign the authorized user sheet to affirm that the above steps have been completed.

- B. Unauthorized Personnel: No unauthorized personnel may enter the BNC clean room facility unless accompanied by an authorized user. All visitors must be briefed on proper safety protocol and must wear appropriate protective eyewear located on the premises.

III. Hazards

- A. Electrical Hazards: Electrical shock or electrocution could result from direct contact with high voltage. Such hazards are typically interlocked by the Evaporator system. High voltage electrodes and conductors are located inside the Evaporator system chassis. In addition, the external Thermal power supply unit has connections behind the power supply chassis. Do not disconnect the external lines. Use normal precautions with external house (110VAC) connections behind the Evaporator chassis.
- B. Chemical: N/A.
- C. Pressure Hazards: Pressurized house gases are used with the Evaporator. Do not disconnect or tamper with gas lines behind the Evaporator. Contact lab management for information.
- D. Other: Bright visible light is generated and emitted through the glass chamber viewing area. Wear protective eye wear if the emitted light is too bright when looking through the chamber.

IV. Hazard Controls

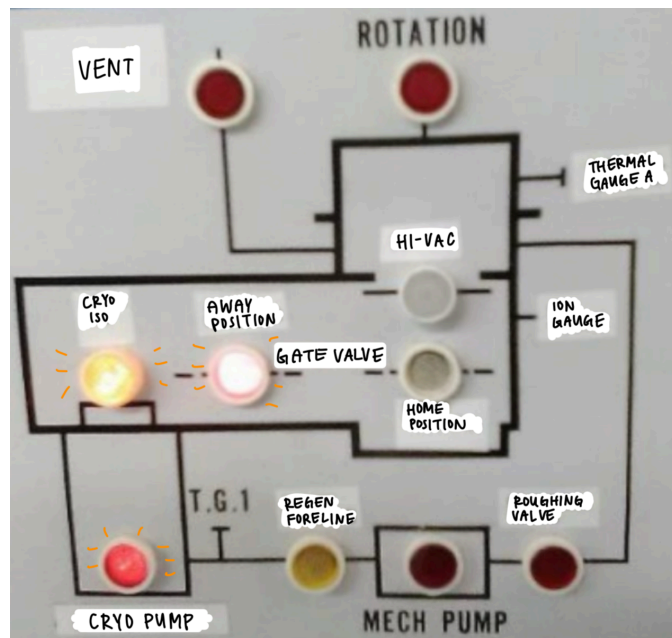
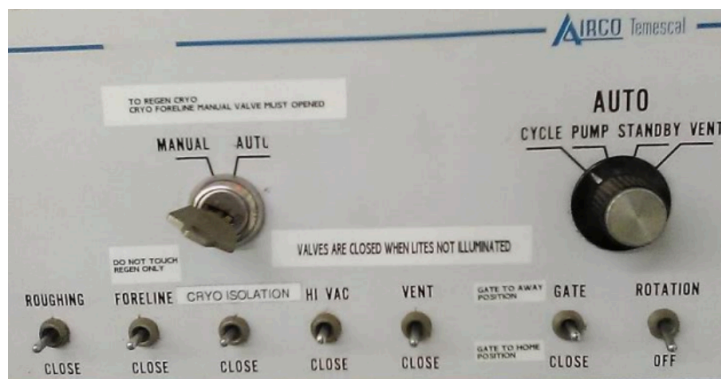
- A. Electrical:
 - 1. Enclosures for protection against the high voltages of the Evaporator power supply may only be removed after the power supply has been unplugged from the outlets and after following the safety procedures outlined in the safety and operations manual provided by the manufacturer.
 - 2. Only qualified personnel may perform all internal maintenance to the Evaporator and more than one user must be present when performing said maintenance.
 - 3. Every portion of the electrical system, including the printed circuit cards, should be assumed to be at dangerous voltage level.
- B. Chemical and Pressure:
 - 1. Enclosures for protection against valves and internal gas plumbing may only be removed after the system has been turned off and gases have been valved off and relieved of line pressure.
 - 2. Only qualified personnel may perform all internal gas maintenance to the Evaporator and more than one user must be present when performing said maintenance.
- C. Other:
 - 1. Proper eye protection and hand protection must be worn at all times in the clean room and while operating the Evaporator.

V. General Maintenance

- A. Chamber must be fully vented to atmospheric pressure with E-Beam crucible exposed.
- B. Move the Gate Valve to HOME position.
- C. Verify the indicator light switching from AWAY to HOME position.
- D. CLOSE the HiVAC valve by pressing down in HiVAC switch.
- E. Verify the HiVAC indicator limit is illuminated and HOME light is OFF.
- F. Open N2 Vent valve by pressing up on the VENT switch.
- G. Verify the Vent indicator light is illuminated.
- H. Allow the chamber several minutes to reach atmosphere. The entire chamber volume including chamber area above the cryo pump is being vented. It may take more time to vent. Note that the Vacuum Gauge B should be 9.9 +9.
- I. After the lid has been released from vacuum, the lid can be opened and must be lifted and opened all the way.
- J. Close N2 Vent valve by pressing down on the VENT switch.
- K. Verify that the Vent indicator is illuminated.
- L. Cryo Pump indicator light remains On.
- M. Verify that the HiVaC valve is closed by pressing down on HiVAC switch.
- N. Move the Gate Valve to AWAY position.
- O. Verify Shutter is in CLOSED position.
- P. The Gate valve will slide for the Gate valve will slide from the position in front of the chamber lid to the area above the cryo valve.
- Q. Detach the planetarium (rotating sample holder) and place it on a clean, lint-free surface.
- R. Remove the quartz crystal microbalance (QCM) sensor carefully from the chamber.
- S. Clean the QCM sensor with the CRV rolling lab vacuum to remove loose debris.
 1. If additional cleaning is needed, wipe the crystal surface gently with isopropyl alcohol (IPA) using a lint-free wipe.
- T. Use the CRV rolling lab vacuum to vacuum the internal assembly of the evaporator.
- U. Open Shutter by pressing up the SHUTTER switch and verify Shutter is OPEN position.
- V. Detach the shutter assembly carefully.
- W. Remove any film buildup or bubbles to achieve a smooth surface using one of the following methods:
 1. Razor blade: Scrape metal residue gently over a trash bin.
 2. Scotch-Brite pad: Lay the shutter flat on a cleanroom paper towel and scrub until smooth.
- X. Replace any aluminum lining as needed.
- Y. Reinstall the shutter assembly securely.
- Z. Move the Gate Valve to HOME position.
- AA. Reinstall the planetarium into the chamber
- BB. Wipe down all external surfaces with isopropyl alcohol and a clean room paper towel to remove any build-up or residue.
- CC. Reinstall the planetarium into the chamber

I. Operation

Section A – Standby Conditions for Evaporator When Not In Active Use:



- A. Main Chamber lid should be closed and under High Vacuum
- B. Valves and related light indicators should be OPEN or illuminated (ON).
 - a. Cryo Iso OPEN
 - b. Cryo pump ON
 - c. Gate Valve in the AWAY position
- C. Valves and related light indicators that should be CLOSE or not illuminated (OFF)
 - a. Roughing Value CLOSED
 - b. Cryo Foreline Valve CLOSED
 - c. Mechanical Pump OFF
 - d. Vent Valve CLOSED
 - e. HiVAC Valve CLOSED
 - f. Gate Valve HOME position should not be illuminated
- D. Shutter should be CLOSED.
- E. Bottom view port shutter in CLOSED position.
- F. High Voltage E-Gun Power Supply and E-gun Controller OFF.
- G. High Voltage E-Gun OFF.

Section B – Preparing the Deposition Sources (adding/replacing source material to crucibles):

- A. Follow steps A-P from V. General Maintenance on page 3
- B. Once deposition sources are accessible, the sources can be changed or loaded. Use source selector on turret control to rotate source turret as needed. Make sure that the source material loaded is the same as the label on the turret control selector instrument panel. The shutters open/close can be controlled on the source turret and shutter control panel.
- C. If a deposition is to be performed after removing/loading new substrates, proceed with Section C. For only Shutdown, go to *Section H*. For only Pumping Down the System for Deposition, go to *Section D*. For After Deposition, to Remove or Exchange the Substrates from Chamber ONLY, go to *Section G*.

Section C – Loading Substrates into the System:

- A. Clean or replace plastic viewing shield located on the chamber lid interior.
- B. Install or remove samples/substrates from the sample holder accordingly. Note that the existing sample holder uses only 6-inch diameter holders or wafers. Kapton tape samples to the 6-inch wafers securely. ONLY USE KAPTON TAPE ON MINIMUM 4 LOCATIONS TO SECURE ONTO 6-INCH WAFER. Else samples may fall off wafer.
- C. Inspect and verify the O-ring on the chamber lid surface is clean, in place, and not damaged.
- D. Gently close the chamber lid and seat it gently on the chamber O-ring.
- E. Proceed to Section D.

Section D – Pumping Down the System for Deposition:

- A. Open the evaporator Cooling water in service corridor.
 - a. Turn blue handle #1 counterclockwise 90°
 - b. Turn black handle #2 counterclockwise 90°
 - c. Turn black handle #3 clockwise 90°
 - d. Turn blue handle #4 counterclockwise 90°
 - e. Turn blue handle #5 clockwise 90°
- B. Ensure the chamber lid is firmly sealed.
- C. Start the mechanical pump by switching the mechanical pump circuit breaker located on the Evaporator chamber left power panel to ON. Verify Mechanical pump indicator is illuminated.
- D. Allow the mechanical pump to stabilize for at least 2-3 minutes.
- E. Move the Gate Valve to HOME position.
- F. Verify that HOME position indicator is illuminated (ON).
- G. Verify the Cryo Iso light is illuminated, if not press up on the switch labelled CRYO ISO. Both Gauge A and Gauge B should drop rapidly to 0.0-0. << not sure if this is the right order for cryo iso
- H. CLOSE the HiVAC valve by pressing down in HiVAC switch.
- I. Verify that HiVAC indicator is illuminated (ON).
- J. Open N2 vent valve by pressing up on the VENT switch.

- K. Verify the Vent indicator light is illuminated.
- L. Load wafers into the planetarium.
- M. CLOSE vent valve by pressing down on the VENT switch.
- N. Start the mechanical pump by switching the mechanical pump circuit breaker located on the Evaporator chamber left power panel to ON. Verify Mechanical pump indicator is illuminated.
- O. Allow the chamber to pump down to 5.0×10^{-2} Torr (50 mTorr) indicated on either Thermal Gauge A or B. Note that Gauge A readings will follow Gauge B. After 50mT, Gauge B will follow Gauge A.
 - a. While waiting, cut three bare wafers in half using a dicing tool and a ruler. Ensure the cuts are along the Y direction and that the halved wafers are as equal in size as possible.
 - b. Place these dice into a clear wafer holder box.
- P. Close roughing valve when Gauge A or B reads less than 5.0×10^{-2} torr. Push down on the switch labelled Roughing Valve to close.
- Q. Verify that the Roughing indicator is not illuminated.
- R. Open the HiVac valve by pressing up the switch labelled HiVAC.
- S. Verify HiVac indicator is not illuminated, and HOME position indicator is illuminated.
- T. Shut off the mechanical pump and verify the MECHANICAL PUMP indicator is not illuminated.
- U. Verify that both VACUUM GAUGE A and VACUUM GAUGE B are displaying 0.0 -0 on the Granville-Phillips vacuum gauge controller.
- V. Turn on the Ion Gauge tube by depressing IG1 ON. The ion gauge should gradually warm up and display vacuum levels and gradually move from 10^{-4} torr towards 10^{-6} torr.
- W. Move the gate valve to the AWAY position by pressing the switch labeled GATE VALVE AWAY position.
- X. Verify the HOME position indicator is not illuminated, and the AWAY indicator is illuminated.
- Y. Proceed to Section E for Alumina Deposition**

Section E – Alumina (Al_2O_3) Deposition:

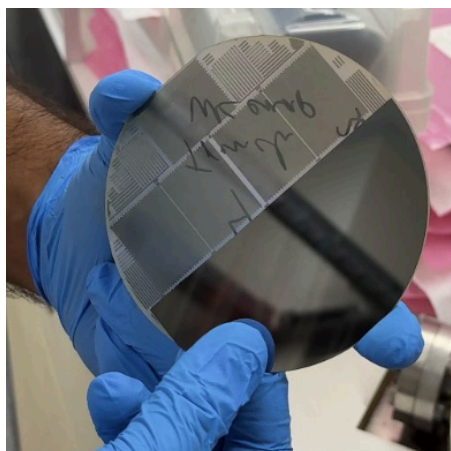
- A. In the “Turret Source and Deposition Mode Selector” section of the control console front panel, turn the knob towards Al_2O_3 (Alumina), with the white arrow on the knob pointing towards the substrate.
- B. Verify the Al_2O_3 (Alumina) indicator “C” is illuminated.
- C. In the “Deposition Controller” section of the control console front panel:
 - a. Press “F6” for program.
 - b. Press “F2” for process directory. → process 4
 - c. Selection Al_2O_3 (Alumina) using the up and down arrows located at the right below the number pad.
 - d. Press “F4” for select active process.
 - e. Verify the active process selection through the digital deposition controller panel
 - i. Top right corner should say which process is selected and which layer to start it is on currently.
 - f. To change or view thickness, press “F5” for process
 - i. press the up arrow once, (thickness is in angstroms) → 0.200 nanometers
 - ii. press enter once done to confirm final thickness
 - g. Press “F6” for process directory.

- h. Press “F4” for select active process again for the updated progress to be added.
 - i. Press “F6” for program.
 - j. Press “F6” for operate.
- D. Turn ON the main power supply crank by flipping it UP.
 - E. Verify the main power indicator is illuminated ON.
 - F. Verify TT-3/6 Control Panel, indicators for Zero, Water, Doors, VAC, and Ready is illuminated ON.
 - G. Turn ON the power supply controller by pressing up on the switch to ON.
 - H. Turn Rotation ON by pressing up on the switch.
 - I. Verify the Rotation indicator is illuminated ON.
 - J. Press “Start” on the “Deposition Controller” section of the control console front panel
 - K. Verify the process by listening in for a click.
 - L. Verify the process once more by observing the V and R of TT-3/6 and the “Voltage/Emission” is illuminated ON.
 - M. The Evaporation Process will go from Soak 1 to Soak 2, then the shutter will OPEN and evaporation starts.
 - N. If the Temperature Display Console Panel shows temperatures from 20.0K to 30.0k, adjust the e-beam position either in the lateral or longitudinal direction.
 - O. Note: deposition might not be smooth, power or temperature can be off << i’m not sure where to add this on the document but yea
 - P. Once the deposition is finished (gone through soak 1, soak 2, evaporation, deposition (steady-state), and idle/ramp down), press “Stop” and then “Reset” on the “Deposition Controller” section of the control console front panel.
 - Q. Verify that the “Voltage/Emission” is illuminated OFF in the TT-3/6 section of the control console front panel.
 - R. Turn rotation OFF by pressing down on the rotation switch.
 - S. To vent the chamber to start the next round of evaporation:
 - a. Turn OFF the power supply controller by pressing down on the switch to OFF.
 - b. Turn OFF the main power supply crank by flipping it DOWN.
 - T. Wait for 2-3 minutes to let the wafers cool down.
 - U. **Proceed to Section F for Titanium Deposition**

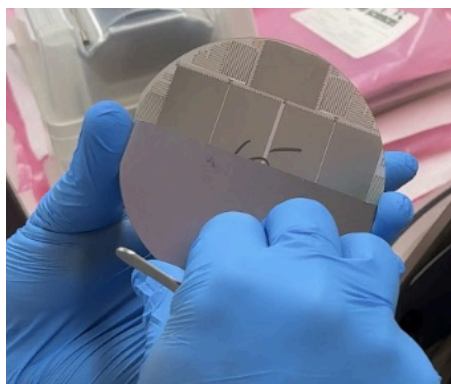
Section F – Titanium (Ti) Deposition:

- A. In the “Deposition Controller” section of the control console front panel:
 - a. Press “F6” for program.
 - b. Press “F2” for process directory. → process 1
 - c. Selection Ti (Titanium) using the up and down arrows located at the right below the number pad.
 - d. Press “F4” for select active process.
 - e. Verify the active process selection through the digital deposition controller panel
 - i. Top right corner should say which process is selected and which layer to start it is on currently.
 - f. To change or view thickness, press “F5” for process

- i. press the up arrow once, (thickness is in angstroms) → 0.300 nanometers
 - ii. press enter once done to confirm final thickness
- g. Press “F6” for process directory.
- h. Press “F4” for select active process again for the updated progress to be added.
- B. In the “Turret Source and Deposition Mode Selector” section of the control console front panel, turn the knob towards Ti (Titanium), with the white arrow on the knob pointing towards the substrate.
- C. Verify the Ti (Titanium) is selected after you heard 2 clicks from the evaporator (since the illuminator is fused → won’t see any illumination)
- D. Follow Section D steps A-K on page 5.
- E. Remove the dice from the wafer holder (refer to part b of step O. in section D)



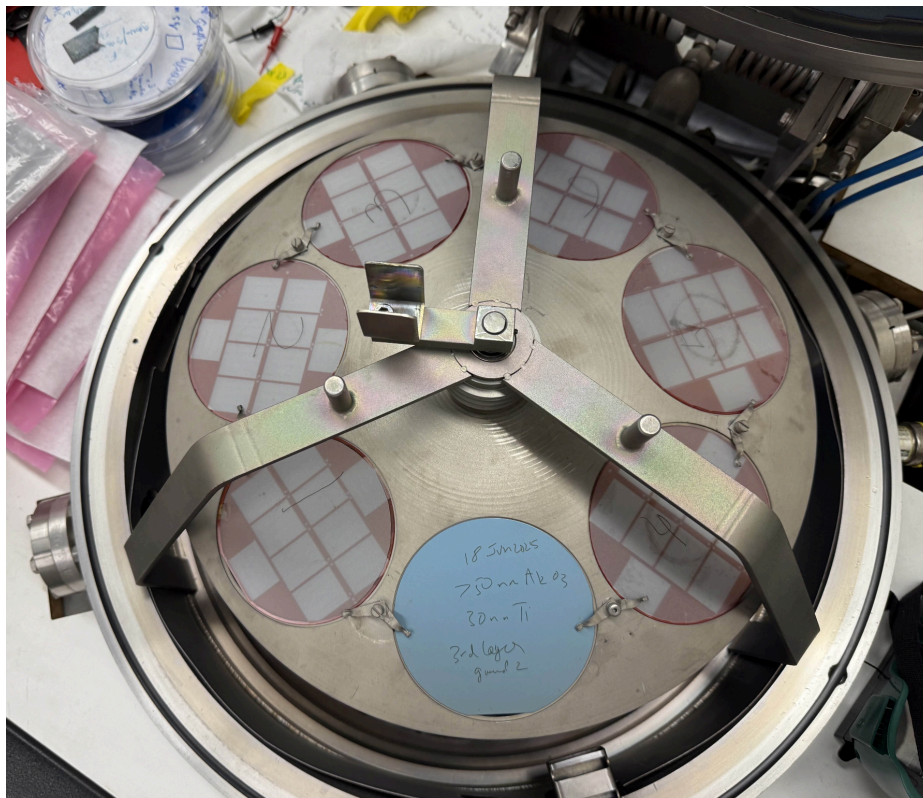
a.



b.

<< need to say which wafer it is, not sure which wafer we added the die on and which one we didnt.. forgot :p

- c. Hang the wafer so the dice face the chamber, with the non-dice side visible. Ensure the dice are completely covered from view, exposing only the wafer side without the dice.



F.

- G. After titanium deposition is done, we will deposit another layer of Alumina, so follow Section D and E once more.

Section G – After deposition, to Remove or Exchange the Substrates from Chamber ONLY:

- A. To reload or change the substrates for additional depositions, the chamber must return to atmosphere. If more source material is to be added, go to SECTION B.
- B. Turn IG1 OFF.
- C. Move the Gate Valve to HOME position.
- D. Verify the indicator light switching from AWAY to HOME.
- E. CLOSE the HiVAC valve by pressing down on the HiVAC switch.
- F. Verify the HiVAC indicator light is illuminated, and HOME light is OFF.
- G. Open N2 vent valve by pressing up on the VENT switch.
- H. Verify the VENT indicator light is illuminated.
- I. Allow the chamber several minutes to reach atmosphere.
- J. After the lid has been released from vacuum, the lid can be opened (Note: It is heavy) and opened all the way.
- K. Close N2 Vent valve by pressing down on the VENT switch.
- L. Verify the VENT indicator is not illuminated.
- M. Install or remove samples/substrates from the sample holder. Note that the existing sample holder uses 6-inch diameter wafers. Kapton tape samples to the 6-inch holder/wafers securely. **ONLY USE KAPTON TAPE ON MINIMUM 4 LOCATIONS TO SECURE ONTO 6-INCH WAFER.**

- N. Inspect and verify the O-ring on the chamber lid surface is clean, in place, and not damaged.
- O. Gently close the chamber lid and seat it gently on the chamber O-ring.
- P. Return to Step D for continued deposition. When the process is finished go to Step G for placing evaporator on Standby and Shutdown.

Section H – All Done and Returning System to Standby Condition:

- A. Ensure that the chamber lid is firmly seated.
- B. Start the mechanical pump by switching the mechanical pump circuit breaker located on the Evaporator chamber left power panel to ON.
- C. Allow the mechanical pump to stabilize for at least 2-3 minutes.
- D. Open the roughing valve by pressing up the switch labelled ROUGHING VALVE.
- E. Verify the ROUGHING indicator is illuminated.
- F. Allow the chamber to pump down to 5.0×10^{-2} Torr (50 mTorr) on either Gauge A or B.
- G. Close ROUGHING VALVE when Gauge A or B reads less than 5.0×10^{-2} torr. Push down on the switch labelled ROUGHING VALVE to close.
- H. Verify that the Roughing indicator is not illuminated.
- I. Check that HiVAC is NOT in AWAY Position. It should be in HOME. See step J.
- J. Verify HiVAC indicator is not illuminated, and HOME position indicator is illuminated.
- K. Move the gate valve to the AWAY position by pressing the switch labeled GATE VALVE AWAY position.
- L. Verify the HOME position indicator is not illuminated, and the AWAY indicator is illuminated.
- M. Verify the CRYO Iso light is illuminated, if not press up on the switch labelled CRYO Iso.
- N. Verify that both VACUUM GAUGE A and VACUUM GAUGE B are displaying 0.0 -0 on the Granville-Phillips vacuum gauge controller. Both Gauge A and B must be less than 50mT before opening CRYO ISO valve.
- O. Shut off the mechanical pump and verify the MECHANICAL PUMP indicator is not illuminated.
- P. The system is now in STANDBY CONDITION.
- Q. Go to the service corridor and turn OFF evaporator Cooling water in service corridor.
 - a. Turn blue handle #1 clockwise 90°
 - b. Turn black handle #2 clockwise 90°
 - c. Turn black handle #3 counterclockwise 90°
 - d. Turn blue handle #4 clockwise 90°
 - e. Turn blue handle #5 counterclockwise 90°
- R. Complete the Logbook information.

Section I – Emergency Procedures:

- A. Evaporator accidents: Notify lab management and PI immediately.
- B. Power outage: if possible, do the following
 - a. Switches on Temescal Front Power Control, all circuit breakers to OFF.
 - b. Switch Evaporator Main Power (gray box) located on room wall behind Instrument to OFF (pull down).
 - c. Note in the logbook power outage date and time.
 - d. Turn OFF all evaporator cooling water in the service corridor.